

MUHAMMAD HASSAN

Unreal Engine Developer (C++ | Blueprints) | Multiplayer (Replication/RPC) | Tools/Plugins | Pixel Streaming | Gaussian Splatting

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PROFESSIONAL SUMMARY

Unreal Engine developer with **4+ years** building gameplay systems, multiplayer features, editor tools, and polished UI/UX in **C++ and Blueprints**. Proven experience delivering **replicated multiplayer + Pixel Streaming** experiences, performance-first optimization, and custom pipelines for **dense spatial capture (point clouds / splats)** and cinematic automation.

CORE SKILLS

- **Unreal Engine 5.x**, C++, Blueprints, UMG, Editor Tools/Plugins
 - **Multiplayer:** Replication, RPCs, Dedicated Servers, **Epic Online Services (EOS)**, Session flow (create/discover/join/leave)
 - **Pixel Streaming** (browser-based gameplay, synced interactions)
 - **Rendering/Tools:** RDG compute passes, GPU readback, export pipelines, logging/profiling/validation
 - **3D Data:** Point grids/XYZRGB sampling, Gaussian Splatting workflows, Nerfstudio-compatible PLY export
 - **Animation:** Animation Blueprints, locomotion, motion matching, Metahuman integration
 - **Optimization:** Profiling/debugging, packaging, level streaming, memory/asset streaming
 - GitHub, Perforce (P4)
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PROFESSIONAL EXPERIENCE

Moshpit Studio — Game Developer | Jul 2021 – Present

- Shipped scalable **gameplay systems** (core mechanics, animation pipelines, modular architecture) designed for fast iteration and long-term maintainability.

- Delivered **replicated multiplayer** plus **Pixel Streaming** for smooth, synchronized browser-based gameplay.
- Built production-quality **UMG + C++/Blueprint tooling**, custom plugins, and documentation supporting reliable **PC + Android** releases.
- Drove **performance-first** profiling/debugging/optimization with cross-discipline collaboration and user testing.
- Built advanced **3D capture tooling** (LiDAR-style scanning, grid/route sampling, accurate mesh bounds) to generate dense, visibility-aware spatial datasets for point-cloud/splat workflows.
- Developed an **AI-driven cinematics system** that plans shots and auto-generates CineCamera actors, animation, and cuts—reducing manual keyframing effort.
- Implemented **data-driven editor UI generation** to auto-build/bind tool panels, reducing repetitive UI work and improving tool consistency.

Skylia Studio (Greece) — Multiplayer Developer | Apr 2025 – Present

- Built a multiplayer game on a **Unreal Engine source build**, integrated with **Epic Online Services (EOS)**.
- Implemented **dedicated server architecture** with session creation, discovery, and client join/leave flows.
- Developed networked gameplay features using **replication + RPCs** for smooth player interaction.
- Designed and implemented **UMG UI/UX** for menus and multiplayer flow screens.

Fiverr — Freelance Graphics & Animation Designer (Level 2 Seller) | May 2019 – Jun 2023

- Delivered **500+** design projects for Twitch/Discord creators and gaming communities.
- Produced motion graphics and creator branding assets (overlays, alerts, transitions, emotes, intros, logos) using After Effects, Photoshop, Illustrator, Premiere Pro.

SELECTED PROJECTS

UnrealEngineTwin | May 2025 – Present

- Built an **EQS-driven autonomous navigation + scanning system** to generate dense, visibility-aware point grids and export **XYZRGB** samples for 3D reconstruction workflows.
- Implemented a real-time **UE5.5 Gaussian Splat capture + export pipeline** using **RDG compute passes + GPU readback**, with correct camera/world reconstruction and stable PIE capture.

- Created a **Nerfstudio-compatible PLY exporter** and added voxel + hierarchical variance-based merging to shrink outputs from **hundreds of MB to tens of MB** (major size reduction while preserving detail).
- Exposed tools via Blueprint-callable interfaces with strong logging, profiling, and validation for reliable large-scale exports.

Virtual Concert Experience (UE5 Technical Systems) | Aug 2022 – Present

- Built a full UE5 virtual concert platform with real-time show control, cinematic tooling, and scalable performance.
- Developed a **Nanite HISM crowd system** and optimized animation/material workflows in C++ + UE materials.
- Implemented runtime clothing switching and **HISM** → **Skeletal Mesh** conversion for interactive crowd characters.
- Created interactive lighting + fireworks sequencing with a custom timeline; added runtime camera recording + shot timeline and VR camera path recording.
- Integrated Pixel Streaming and optimized level streaming/packaging/memory/asset streaming to reduce load friction in large environments.

Multiplayer Squads Game (Invasion) | Mar 2025 – Present

- Built EOS-based multiplayer systems: sessions, squad creation, server browsing, achievements, and synchronized gameplay replication.

EDUCATION

BS, Information Technology — Punjab University College of Information Technology (PUCIT) | 2018 – 2022